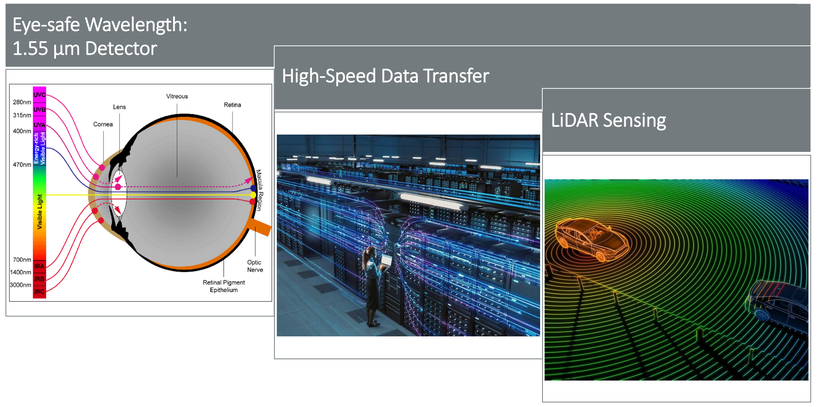
Heterogeneously Integrated GaAsSb/Si Photodiodes with an Active Interface

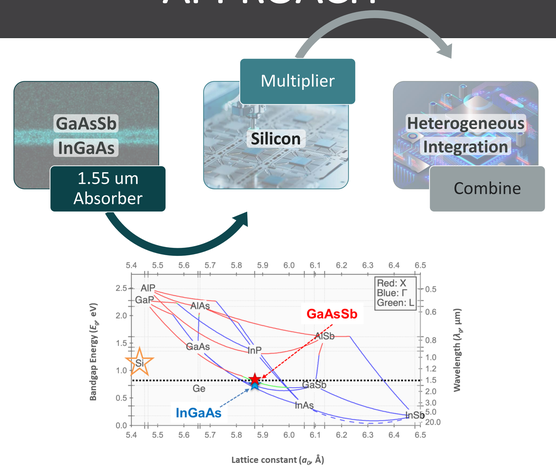
M. Muduli, Y. Xia, S. Lee, N. Gajowski, C. Chae, S. Rajan, J. Hwang, S. Arafin, and **S. Krishna***

THE OHIO STATE UNIVERSITY
COLLEGE OF ENGINEERING

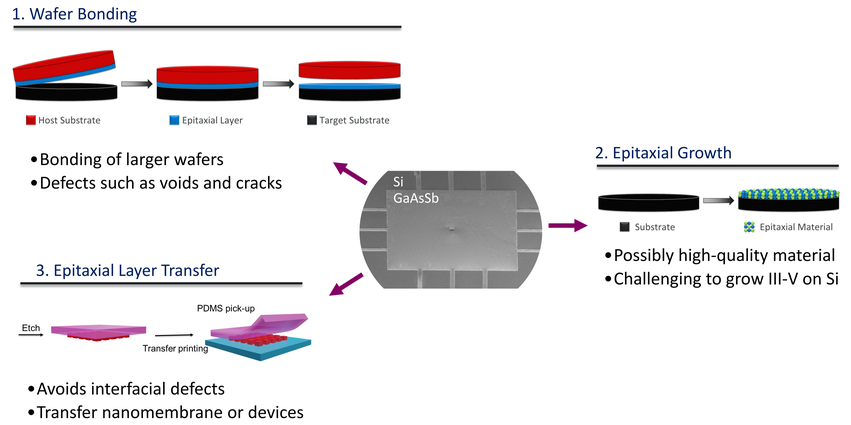
MOTIVATION



APPROACH

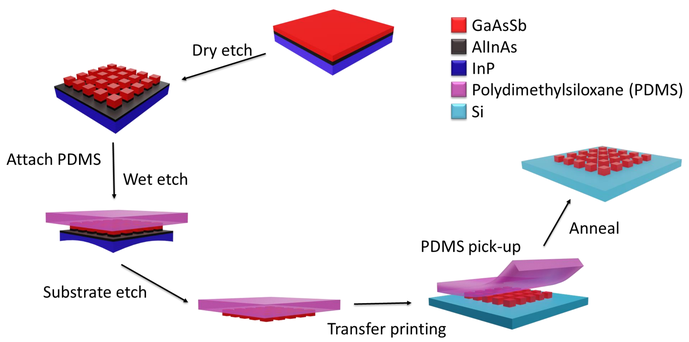


METHODOLOGY



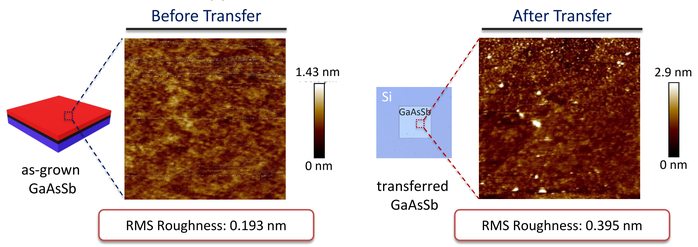
TRANSFER PROCESS

Macro – Transfer Printing



Surface Analysis

Atomic Force Microscopy

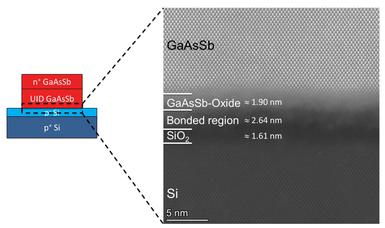


Membrane quality is maintained after the transfer process.

CHARACTERIZATION

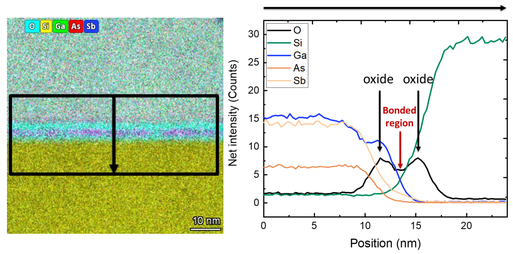
Interface Analysis

Transmission Electron Microscopy



TEM confirms Defect Free Interface.

Energy Dispersive X-Ray Spectroscopy

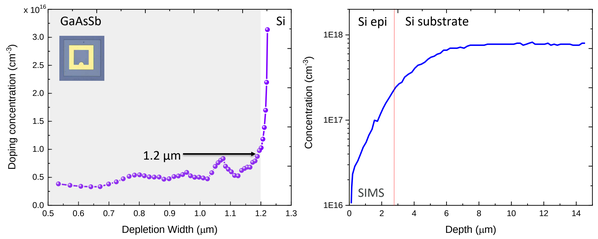


EDS confirms the presence of Gallium oxide on GaAsSb side.

"Bonded Region" is an amorphous mixture of Ga, Si, and O.

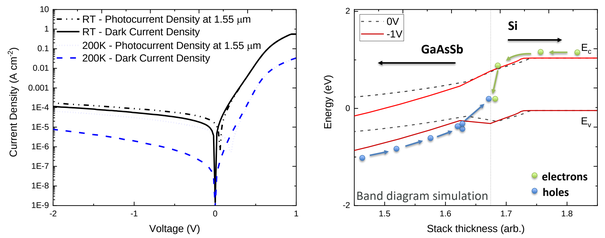
Device Characterization

Capacitance-Voltage Analysis



Si starts depleting before GaAsSb is fully depleted

Current-Voltage Measurements



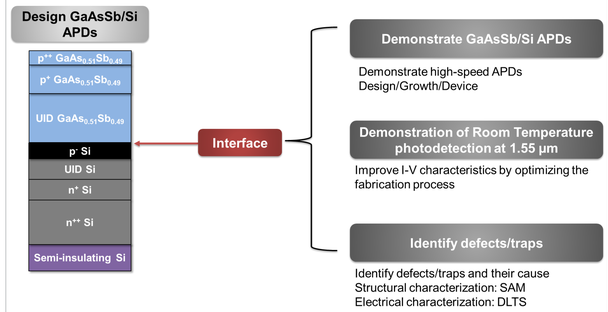
GaAsSb/Si heterostructure responds to 1.55 µm.

The first step to high-speed detectors is complete!

CONCLUSION

1. Successfully integrated GaAsSb with Si and demonstrated a defect-free interface.
2. Demonstrated GaAsSb/Si PIN diodes with response to 1.55 µm wavelength.

Future work



REFERENCES

[1] Abbasi, H.N., Lee, S., Jung, H., Gajowski, N., Lu, Y., Wang, Y., Kim, D., Zhou, J., Gong, J., Chae, C. and Hwang, J., 2024. Structural and electrical properties of grafted Si/GaAsSb heterojunction. *Applied Physics Letters*, *125*(10).
[2] Muduli, M., Xia Y., Lee, S., Gajowski, N., Chae, C., Rajan, S., Hwang, J., Arafin, S., Krishna, S., 2024. arXiv:2407.17607v2
[3] Nallamothu, N.S., Xia, Y., Nikor, S.S.S., Jung, H., Gajowski, N., Lee, S., Arafin, S., Krishna, S. and Reano, R.M., 2023, October. Direct Bonding of GaAsSb to Silicon for High-Speed Avalanche Photodiodes. In *Laser Science* (pp. JM7A-105). Optica Publishing Group.
[4] Xia, Y., Nikor, S.S.S., Nallamothu, N.S., Adams, R.L., Jung, H., Gajowski, N., Lee, S., Reano, R.M., Krishna, S., Ringel, S. and Arafin, S., 2023, November. Fabrication of Si/GaAs 0.51 Sb 0.49 Heterostructure Diodes via Transfer Printing. In *2023 IEEE Photonics Conference (IPC)* (pp. 1-2). IEEE.

KIND
Krishna INfrared Detector Laboratory

***Prof. Sanjay Krishna (Advisor)**
George R Smith Professor of Engineering
Electrical and Computer Engineering

Abbreviations:-
- RT: Room Temperature
- Epi: Epitaxially grown layer
- AFM: Atomic Force Microscopy
- TEM: Transmission electron Microscopy
- EDS: Energy Dispersive X-Ray Spectroscopy
- SIMS: Secondary Ion Mass Spectroscopy
- XRD: X-Ray Diffraction
- APD: Avalanche Photodiode
- SAM: Scanning Acoustic Microscopy
- DLTS: Deep Level Transient Spectroscopy

Funding:
This work was funded by Intel's CAFÉ (Center for Advanced Semiconductor Fabrication Research and Education) program for Ohio.